

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)
16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)

21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I kuruba . Anji(192412682) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Kuruba . Anji

RUBRICS FOR CALCULATION:

Article Review

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Edge Computing over Satellite Networks for Future Wireless Systems

One-Page Research Write-Up

Introduction

Future wireless systems, particularly 6G networks, are expected to provide communication and computing services to users regardless of location. However, many remote, rural, oceanic and disaster-affected regions have limited terrestrial infrastructure. Satellite networks, especially Low Earth Orbit (LEO) constellations, can extend wireless coverage to these areas, while edge computing can process data closer to users instead of sending every task to distant cloud servers. The integration of satellite communication with edge computing therefore creates a promising architecture for low-latency and ubiquitous wireless services. Cao et al. proposed an edge-assisted multi-layer MEC architecture in which LEO satellites act as additional computing nodes alongside terrestrial base stations, enabling computation-intensive tasks to be processed closer to end users.

Architecture and Working

In a satellite-assisted edge computing system, a user device first generates a computation task such as image analysis, sensing, autonomous control, augmented reality or industrial monitoring. The task may be executed locally, offloaded to a nearby terrestrial MEC server, or transmitted through a satellite to an edge server located in the satellite or in another layer of the satellite-terrestrial network. A task-offloading controller selects the most suitable computing location by considering channel conditions, available processing capacity, transmission delay, and energy consumption. LEO satellites are particularly suitable because their lower altitude offers lower propagation delay than higher-orbit satellite systems.

Benefits for Future Wireless Systems

The main advantage of satellite edge computing is wider access to computing services without depending entirely on terrestrial infrastructure. Moving computation toward the network edge can reduce the amount of raw data transferred to distant cloud facilities, lowering communication delay and potentially reducing device energy consumption. In 6G networks, this approach can support applications such as remote healthcare, disaster response, smart agriculture, maritime connectivity, connected vehicles, drone operations and large-scale Internet of Things systems. Multi-layer coordination can also balance workloads between satellites and ground MEC servers, improving resource utilization and service continuity.

Challenges and Future Scope

Satellite edge computing still faces important technical challenges. LEO satellites move rapidly, causing frequent changes in topology, link quality and available computing resources. Satellite payloads also have limited energy, processing power and storage compared with terrestrial data centers. Efficient task scheduling, inter-satellite cooperation, beam management, caching and resource allocation are therefore necessary. Future research can use artificial intelligence and deep reinforcement learning to predict traffic demand, select offloading decisions dynamically, and coordinate computing resources across satellite and terrestrial layers. Secure orchestration, energy-aware satellite computing and integrated sensing-and-communication services are also promising directions for 6G.

Conclusion

Edge computing over satellite networks can transform satellites from communication-only infrastructure into distributed computing platforms. The combination of LEO connectivity, terrestrial MEC and intelligent resource management can provide ubiquitous, responsive and resilient services for future wireless systems. The research by Cao et al. demonstrates that multi-layer offloading can achieve lower computing latency and energy dissipation than single-layer alternatives, highlighting the potential of satellite-edge integration as an important component of 6G networks.

Reference

X. Cao, B. Yang, Y. Shen, C. Yuen, Y. Zhang, Z. Han, H. V. Poor, and L. Hanzo, "Edge-Assisted Multi-Layer Offloading Optimization of LEO Satellite-Terrestrial Integrated Networks," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 2, pp. 381–398, Feb. 2023, doi: 10.1109/JSAC.2022.3227032.

REFERENCE PAPER

Multiple Access in the Era of Distributed Computing and Edge Intelligence

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and George K. Karagiannidis, *Fellow, IEEE*

Abstract—This paper focuses on the latest research and innovations in fundamental next-generation multiple access techniques and the coexistence with other key technologies for the sixth generation (6G) of wireless networks. In more detail, we first examine multi-access edge computing, which is critical to meeting the growing demand for data processing and computational capacity at the edge of the network, as well as network slicing. We then explore over-the-air computing, which is considered to be an approach that provides fast and efficient computation of various functions. We also explore semantic communications, identified as an effective way to improve communication systems by focusing on the exchange of meaningful information, thus minimizing unnecessary data and increasing efficiency. The interrelationship between machine learning (ML) and multiple access technologies is also reviewed, with an emphasis on federated learning, federated distillation, split learning, reinforcement learning, and the development of ML-based multiple access protocols. Finally, the concept of digital twinning and its role in network management is discussed, highlighting how virtual replication of physical networks can lead to improvements in network efficiency and reliability.

Index Terms—Next-generation multiple access (NGMA), multi-access edge computing (MEC), over-the-air computing (OTA), semantic communications, machine learning, digital twinning

I. INTRODUCTION

THE next generation of wireless communication networks is expected to provide intelligence, limitless connectivity and full synchronization of the physical and digital worlds, paving the way for the development of novel applications both indoors and outdoors. Specifically, the key performance indicators (KPIs) expectations of academia and industry are converging toward an ever-increasing deployment of machine-type nodes in 2D/3D service areas (10 Gbps/m³), and the need for even higher reliability (10⁻⁹ frame error rate), lower latency (0.1 ms), higher data rates (1 Tbps), higher energy efficiency (1 pJ/bit) [1]–[3].

However, except achieving higher data rates, lower latency, and more connected devices, future wireless networks can be

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Project Number: 15642).seen as a distributed computing platform and, thus, purely communication metrics are not sufficient to describe their performance. In more detail, leveraging wireless networks as a flexible and scalable computing platform fundamentally alters the design and optimization of multiple access schemes. The primary objective shifts towards efficiently processing users’ data, achieved either independently at any available processing unit or by conducting calculations using the datasets from multiple users as input. In addition, due to the complex nature of future wireless networks, they are expected to integrate native artificial intelligence (AI), which will facilitate the optimal use of communication and distributed computing resources, as well as the provision of services that are based on distributed artificial intelligence.

A. Background on multiple access

To meet the stringent requirements of next-generation wireless networks, a key challenge is the development of advanced multiple access schemes, known as next-generation multiple access (NGMA). NGMA aims to efficiently and intelligently connect numerous users and devices using available wireless resources. Historically, multiple access schemes have played a critical role in wireless communications. These schemes are broadly divided into two types: orthogonal and non-orthogonal transmission strategies [4].